



Broncer — Indian Cinema Recommender

TF.JS

Content-based ML model · Train locally · Save · Reload · Predict



Build & train embedding model

content-based filtering

Encodes 40 Indian movies as feature vectors (genre, era, language, mood, rating). Trains a dense autoencoder to learn a 6-dimensional latent space — similar movies cluster together.

```
// Each movie → 14-dim feature vector
// [genre_onehot×6, language×3, era×2, mood×2, rating_norm]
const encoder = tf.sequential({
  layers: [
    tf.layers.dense({ inputShape:[14], units:10, activation:'relu' }),
    tf.layers.dense({ units:6 }), // latent space
    tf.layers.dense({ units:10, activation:'relu' }),
    tf.layers.dense({ units:14, activation:'sigmoid' }) // reconstruct
  ]
});
model.compile({ optimizer:'adam', loss:'meanSquaredError' });
```

Epoch 200 / 200

loss 0.02987

MOVIES

40

FINAL LOSS

0.02987

LATENT DIM

6

✓ Trained

✓ Trained on 40 Indian movies · final loss: 0.02987



Save model locally

indexedDB://

Persists the trained autoencoder and all 40 movie embeddings to browser storage — survives page reloads.

```
await model.save('indexeddb://broncer-movies');

// Save embeddings + movie catalog to localStorage
localStorage.setItem('broncer-movie-data', JSON.stringify({
```

```
embeddings, movies
))) ;
```

✓ Saved

↓ Download

✓ Model + embeddings saved to IndexedDB & localStorage



Reload saved model

tf.loadLayersModel

Restores the full autoencoder and movie embeddings from storage – ready for inference without retraining.

```
const loaded = await tf.loadLayersModel('indexeddb://broncer-movies');
const data = JSON.parse(localStorage.getItem('broncer-movie-data'));
```

```
[21:36:38] Connecting to IndexedDB...
[21:36:38] Autoencoder weights restored (14→10→6→10→14)
[21:36:38] 40 movie embeddings loaded from localStorage
[21:36:39] Model ready for cosine similarity search ✓
```

✓ Reloaded

✓ Reloaded. Select movies below to get recommendations.



Get recommendations

cosine similarity

Select movies you like – the model encodes them, averages the embeddings, then finds the closest neighbors in latent space.

PICK 1-5 MOVIES YOU ENJOY

Sholay

Action · Hindi

Deewar

Drama · Hindi

Mughal-E-Azam

Drama · Hindi

Guide

Drama · Hindi

Anand

Drama · Hindi

Dilwale Dulhania Le Jayenge

Romance · Hindi

Lagaan

Drama · Hindi

Dil Chahta Hai

Comedy · Hindi

Gangs of Wasseypur

Thriller · Hindi

3 Idiots

Comedy · Hindi

Taare Zameen Par

Drama · Hindi

Queen

Drama · Hindi

Dangal

BioPic · Hindi

Andhadhun

Thriller · Hindi

Gully Boy

Drama · Hindi

Zindagi Na Milegi Dobaara

Comedy · Hindi

PK
Comedy · Hindi

Kabir Singh
Romance · Hindi

Sanju
BioPic · Hindi

URI: The Surgical Strike
Action · Hindi

Nayakan
Drama · Tamil

Roja
Romance · Tamil

Anniyan
Thriller · Tamil

Vikram Vedha
Thriller · Tamil

Pariyerum Perumal
Drama · Tamil

Master
Action · Tamil

Jai Bhim
Drama · Tamil

Soorarai Pottru
BioPic · Tamil

Magadheera
Action · Telugu

Eega
Action · Telugu

Arjun Reddy
Romance · Telugu

Baahubali: The Beginning
Action · Telugu

Baahubali 2
Action · Telugu

RRR
Action · Telugu

Pushpa: The Rise
Action · Telugu

Drishyam
Thriller · Hindi

Tumbbad
Thriller · Hindi

Padmaavat
Drama · Hindi

Shershaah
BioPic · Hindi

Kantara
Drama · Hindi

◆ Recommend

Clear

RECOMMENDED FOR YOU

- 1

URI: The Surgical Strike

2019 · Hindi · ★ 8.2

Action

98%
- 2

Soorarai Pottru

2020 · Tamil · ★ 8.8

BioPic

97%
- 3

Dangal

2016 · Hindi · ★ 8.4

BioPic

96%
- 4

Shershaah

2021 · Hindi · ★ 8.4

BioPic

96%
- 5

Andhadhun

2018 · Hindi · ★ 8.3

Thriller

96%
- 6

Gangs of Wasseypur

2012 · Hindi · ★ 8.2

96%